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Assessing water security and governance complexity in the Owabi river catchment in Ghana

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Abstract

Water resource management is a pressing challenge in developing regions, including Ghana, where climate change effects and rapid urbanisation, deforestation, and pollution increasingly threaten water availability and quality. This study examined (i) water security conditions and (ii) water governance and management practices in the Owabi River Catchment in Kumasi, Ghana. A mixed-methods approach was employed, comprising a household survey of 1015 participants to assess water access, quality, and use patterns. Furthermore, key informant interviews were conducted to examine governance arrangements and institutional practices. Binary logistic regression was done to analyse determinants of effective water management. The study population reflected the demographic structure of the catchment, with participants drawn from urban, peri-urban, and rural communities and representing diverse age groups, education levels, occupations, and household sizes. Findings reveal water security challenges, including widespread river pollution, declining water quality, and health risks linked to poor domestic wastewater disposal and rising negative land-use pressures. Governance-related challenges were characterised by weak regulatory enforcement, limited institutional coordination, and predominantly passive forms of community participation in water management decision-making. Statistical analysis revealed that community participation ($OR=2.042$, $p<0.001$), water treatment practices ($OR=3.926$, $p<0.001$), and pollution control measures ($OR=15.75$, $p<0.001$) significantly influenced the effectiveness of water resource management. Strengthening participatory governance, improving pollution control, and integrating technological solutions such as Internet of Things-based monitoring and green infrastructure could substantially enhance water sustainability in the Owabi River Catchment.

Keywords Water management, Pollution control, Community participation, Remote sensing, Owabi river catchment, Ghana

1 Introduction

Water is an essential natural resource that supports human survival, economic growth, and ecological sustainability (UNESCO World Water Assessment Programme [WWAP], 2024). However, increasing global demand for freshwater, estimated to be rising at just



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under 1% per year, driven by rapid population growth of approximately 0.9% globally in 2025, industrial expansion, and climate change, has placed immense pressure on water (UNESCO World Water Assessment Programme [WWAP], 2024; U.S. Census Bureau, 2024). For example, the WHO and UNICEF progress report on household drinking water and sanitation during the World Water Week 2025 revealed that, 1 in 4 – or 2.1 billion people globally still lack access to safely managed drinking water, including 106 million who drink directly from untreated surface sources [62]. Furthermore, water scarcity is expected to intensify in many regions, particularly in rural communities of developing countries with fragile contexts [19, 56]. The growing competition for water among domestic, agricultural, and industrial users further exacerbates this crisis, necessitating effective water management approaches for long-term sustainability [21]. However, Integrated Water Resources Management (IWRM) has emerged as an essential framework for balancing social, economic, and environmental demands on water systems, despite its limitations [38, 51]. IWRM promotes coordinated development and management of water resources while enhancing ecosystem resilience [33, 42].

The United Nations Sustainable Development Goal (SDG) 6 further underscores the urgency to ensure universal access to water and sanitation by 2030 [58, 59]. However, despite this and the goals of many other international development institutions, as well as national and local governments, the implementation of water management and governance principles remains ineffective [51]. Developing countries struggle with fragmented policies, weak enforcement mechanisms, and inadequate financial resources, limiting the effectiveness of water management strategies [38, 52].

A considerable amount of literature has been published on water management challenges in Africa caused by social, economic, political and natural factors [2, 28, 52, 28]. Furthermore, rivers and other water bodies that serve as crucial lifelines for socioeconomic development are increasingly threatened by sedimentation, industrial waste, and encroachment [39, 41, 54]. The Volta, Pra, and Owabi Rivers in Ghana, for example, face significant risks from land use changes and weak regulatory enforcement [45, 41]. These pressures underscore the need for context-specific and community-driven solutions to enhance water security.¹ Importantly, the Owabi River catchment in Southern Ghana, which supports diverse uses, including domestic water supply, agriculture, and biodiversity conservation, is particularly over-exploited due to increasing population, which has grown by approximately 20% over the period 2015–2025, reflecting Ghana's average annual population growth rate of about 2.1% [24, 7], with increasing demands for natural resources [7, 19]. The Owabi Reservoir, for example, plays a critical role in urban water security as it supplies water to Kumasi, Ghana's second-largest city [19]. Additionally, the catchment hosts the Owabi Wildlife Sanctuary, an ecologically significant area that supports various plant and animal species [19]. However, rapid urbanisation, agricultural expansion, and deforestation have degraded water quality and reduced water availability [7, 19, 41]. Similarly, governance issues, policy fragmentation, and inadequate community participation further complicate efforts to implement effective water management practices [16, 38].

¹ This refers to having enough clean water of acceptable quality to meet the needs of the individuals, ecosystems and the economy while also ensuring protection from water related risks. It aims to make the most of water's benefit for humans and ecosystems.

The degradation of the Owabi River catchment highlights the interconnected nature of environmental, socio-economic, and governance challenges [7, 19]. Agricultural runoff contributes excess nutrients and pesticides to the river, leading to eutrophication and water quality deterioration [19, 41]. Industrial and domestic waste, often discharged untreated, increases treatment costs and poses severe risks to public health [41]. Additionally, deforestation and unsustainable land uses disrupt the hydrological cycle, reducing groundwater recharge and increasing water scarcity [7, 50]. Similarly, governance weaknesses, particularly fragmented institutional mandates, weak enforcement of environmental regulations, and limited community participation, remain a critical barrier to sustainable water management in the Owabi River Catchment [19, 38, 52]. Many water policies in Ghana lack coherence, with overlapping mandates among different government agencies leading to inefficiencies in decision-making and enforcement [19, 52]. Moreover, community participation in water governance remains limited, despite evidence that inclusive decision-making leads to more effective management outcomes [34].

Despite a growing body of literature on water management and governance, several key gaps remain. For example, few studies comprehensively integrate biophysical water quality assessments with socio-economic and governance factors in the Owabi River Catchment [39, 41, 42]. In addition, there is a lack of empirical, community-based data that quantifies how pollution, accessibility, and health outcomes interact to shape water governance effectiveness, and limited attention has been paid to the statistical relationships between community participation, pollution control, and sustainable management within a catchment-level context [7, 17, 19, 39, 42]. The central question in this study is: how do governance structures influence water management in the Owabi River Catchment? The more specific question is: how do socio-economic factors affect accessibility and the quality of water? The study, therefore, examined (i) water security conditions and (ii) water governance and management practices in the Owabi River Catchment in Kumasi, Ghana.

2 Water security and management of drainage basins: a literature review

River basins are important water sources that ensure groundwater recharge and a steady supply of water for different uses. Studies show that river basins are useful for the functioning of the hydrological system [32, 39]. Thus, proper management of drainage basins is critical to ensuring the health of ecosystems, economies, and communities [26, 15]. However, river basins, which are water sources, particularly for rural communities, are over-exploited, coupled with the negative effects of climate change, on water resources [50]. Globally, water security² faces significant challenges from rising population, increasing urbanisation, growing industrial demands, intensification and extensification of agriculture, rising emissions of waste and pollution, and the effects of climate change [27, 42, 54]. For example, about 40% of the world's population experiences water scarcity, while floods affect millions annually [59]. Despite several approaches, centralised and top-down approach [2, 52], the community-based and bottom-up approach [2, 51], polycentric management [15], integrated management [38, 34], among others,

²The capacity of a population to safeguard sustainable access to adequate quantities of acceptable quality water for sustaining livelihoods, human well-being, and socio-economic development, for ensuring protection against water-borne pollution and water-related disasters, and for preserving ecosystems in a climate of peace and political stability (<https://www.unwater.org/publications/what-water-security-infographic>).

water management and governance challenges persist. Recent evidence suggests that the effects of climate change are likely to increase the stress on water resources in different regions of the world at various intensities [18], & Tortajada, 2019). Furthermore, inadequate climate data has hindered measures to quantify the social and economic impact of climate-related changes on water resources [9].

Water security varies greatly depending on climatic, geographic, and socio-economic conditions (UN-Water, 2013). Some countries in Sub-Saharan Africa face persistent water scarcity due to low rainfall and high evaporation rates. Although strategies such as rainwater harvesting, groundwater management, and improved irrigation practices are essential for mitigating these challenges, they are still not effective. Tropical regions, on the other hand, such as Central Africa and Southeast Asia, frequently struggle with waterlogging and floods caused by excessive rainfall.

Ghana's institutional framework for water governance is based on the principles of decentralization and participatory management [4]. However, its effectiveness at the local level remains constrained by fragmented and overlapping institutional mandates, weak inter-agency coordination, and limited enforcement capacity [19, 52]. Consequently, this study adopts a theoretical perspective to better understand the interactions among institutions, stakeholders, and water users within the Owabi catchment. This allows for a more nuanced examination of governance complexity, power relations, and participation dynamics that are not sufficiently captured by Ghana's formal water governance framework alone [11, 47, 52]. A relevant guiding model is the Integrated Water Resources Management (IWRM) framework, which encourages the coordinated development and management of water, land, and related resources to maximize economic and social welfare without damaging ecosystems. Ghana's water policy aligns with IWRM principles, highlighting decentralization, stakeholder participation, and sustainability [60]. Furthermore, traditional management systems have proven to be effective in environmental conservation in Ghana. These authorities, for example, hold significant influence, particularly in rural and peri-urban areas, where they oversee land and water resource allocation [3]. Noteworthy is that Ghana's customary laws have historically governed the use of natural resources, emphasizing conservation and community ownership [1]. However, in contemporary governance frameworks, this traditional role often conflicts with state-led policies, creating overlaps and inefficiencies [45]. Also, implementing IWRM at the local level remains difficult, mainly due to weak inter-agency coordination and limited community engagement [19]. Furthermore, the notion of stakeholder engagement examined through the lens of participation theory, illuminates Arnstein's Ladder of Participation, which categorises citizen involvement from non-participation to degrees of tokenism and citizen power (Arnstein, 1969). In the context of the Owabi catchment, for example, participation is often consultative, with community members having limited influence on policy or project outcomes [3]. This limits the effectiveness of participatory mechanisms embedded in the country's water governance strategy. However, The National Community Water and Sanitation Programme (NCWSP) promotes local ownership, with the Community Water and Sanitation Agency (CWSA) guiding service delivery in rural areas [4]. Additionally, the concept of polycentric governance describes the overlapping/conflicting mandates and fragmented / unclear/ obfuscated roles and responsibilities among water-related institutions such as the Water Resources Commission (WRC), Ghana Water Company Limited (GWCL),

Environmental Protection Agency (EPA), and traditional authorities (Fig. 1). Polycentric governance emphasizes the coexistence of multiple, autonomous decision-making centres operating at different scales, and suggests that such systems can foster resilience, flexibility, and improved accountability in resource governance [11, 15, 46]. Within the Owabi context, this complexity presents both opportunities for localized solutions and challenges in achieving coordinated governance outcomes.

2.1 Water security and local management practices in the Owabi river catchment

Water security challenges are closely linked to both environmental pressures and human activities in the Owabi River Catchment [7, 27]. In addition to water pollution within the community, extensive irrigation for diverse agricultural purposes is common, further intensifying pressure on available water resources [27]. These pressures have implications for water availability, quality, and ecosystem sustainability, particularly in a catchment that supports domestic water supply, agriculture, and biodiversity conservation [7, 19]. However, local farmers and institutions have devised various strategies to manage water resources and enhance sustainability [27]. Integrated water management practices that bring together different water users and uses, coupled with environmental conservation measures, are promoted by municipal authorities and Village Development Organisations (VDOs) as part of local-level water management efforts [27, 45]. Alongside these formal arrangements, traditional systems of water conservation continue to play an important role in regulating access to water resources and promoting stewardship at the community level [45]. Although these indigenous practices have demonstrated effectiveness in managing shared water resources, their integration with modern statutory governance frameworks remains limited, constraining their broader influence on catchment-level planning and enforcement [53, 45].

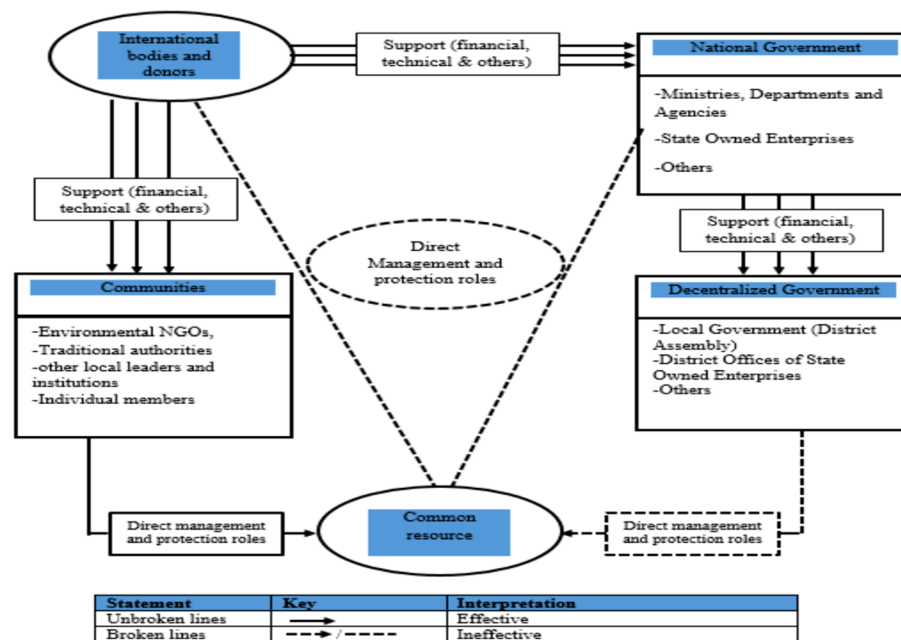


Fig. 1 Conceptual framework for assessing water governance and community participation in the Owabi River Catchment. Source: (Fieldwork, 2022)

Empirical studies conducted in the Owabi River catchment, consistently, document water quality degradation and highlight the growing pressures on water resources arising from land-use change, agricultural activities, and pollution [27, 41, 45]. This underscores the need to examine not only biophysical water security challenges but also the governance arrangements that shape how water resources are managed at the catchment scale [19, 52].

2.2 Water governance and its contextual management in the Owabi river catchment

Water governance in Ghana is formally grounded in the principles of Integrated Water Resources Management (IWRM), decentralisation, and stakeholder participation, with responsibilities distributed among institutions such as the Water Resources Commission, Ghana Water Company Limited, the Environmental Protection Agency, and Metropolitan, Municipal and District Assemblies (Fig. 1). In theory, this governance framework is intended to promote coordinated planning, sustainable allocation of water resources, and inclusive decision-making across national, regional, and local governance levels.

In practice, however, the effectiveness of Ghana's water governance framework at the catchment level remains constrained, particularly in rapidly urbanising and environmentally sensitive areas [19]. These governance weaknesses are evident in the Owabi River catchment, where multiple state and non-state actors with overlapping responsibilities operate within a complex institutional landscape, limiting effective pollution control, land-use regulation, and meaningful community engagement in water management processes [43, 45].

Given these limitations, this study adopts an integrated governance framework that combines IWRM principles with participatory and polycentric governance perspectives to better capture the realities of water management in the Owabi River catchment [11, 47, 52]. The integration of participatory governance emphasises the role of community involvement in decision-making, while polycentric governance recognises the existence of multiple decision-making centres operating across different scales, which is particularly relevant in contexts characterised by institutional overlap and governance complexity [11, 47]. This combined framework provides an appropriate analytical lens for examining how institutional interactions, participation dynamics, and governance structures shape water security outcomes within the Owabi River catchment [11, 19].

3 Research methodology

3.1 Description of the study site

The Owabi River catchment is located between latitudes 6°32' and 6°45' North and longitudes 1°36' and 2°00' West of the Greenwich Meridian (Fig. 2) [7]. This water catchment covers approximately 69 km² and serves as a critical water source for Kumasi and its surrounding communities [19]. Importantly, the catchment is a Ramsar-designated area characterised by semi-deciduous woodland vegetation, undulating topography, and a semi-equatorial climate marked by a double maximum rainfall regime ranging between 170 and 185 cm per annum (Ramsar [7, 48]). The major rainy season occurs between March and July, while the minor rainy season spans from August to November. Temperature within the catchment is relatively uniform throughout the year, ranging between about 27 °C in August and 31 °C in March (GMet, 2020). Relative humidity

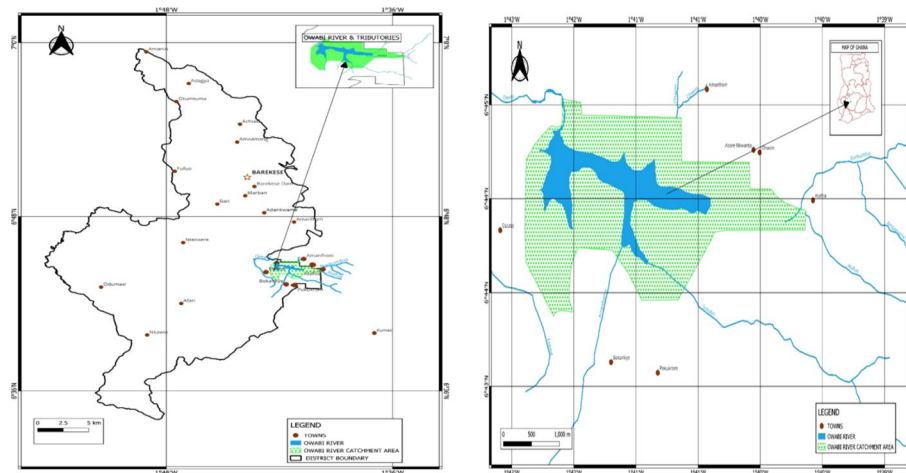


Fig. 2 Map showing the Owabi River in the District Context and Catchment Area. Source: (Authors, 2024)

in the area is generally high, averaging approximately 93%, reflecting the humid tropical climatic conditions of the catchment (GMet, 2020).

Administratively, the Owabi River catchment lies within the Atwima Nwabiagya Municipality in the Ashanti Region of Ghana, which is predominantly urban, with about 68% of residents living in urban and peri-urban areas according to the 2021 Population and Housing Census [24]. Recent spatial analysis conducted by the Atwima Nwabiagya Municipal Assembly's Physical Planning Department indicates that approximately 71.8% of land within the Owabi catchment had been converted into residential development by 2022 (Atwima Nwabiagya Municipal Assembly, 2022). This rapid urban expansion has intensified pressure on water resources through increased abstraction, land-use change, and pollution, thereby heightening water security risks within the catchment [7, 19]. Thus, understanding how existing governance arrangements respond to these pressures is critical for improving water management effectiveness and ensuring sustainable water security outcomes in the Owabi River Catchment [19].

3.2 Data collection

This study employed a mixed research approach using administered questionnaires, direct field observations, and key informant interviews to assess water security and governance in the Owabi River Catchment. Data were collected from 1,015 households, comprising adult men and women aged 18 years and above, representing urban, peri-urban, and rural communities within the catchment [19, 27]. The participants reflected diverse socio-economic backgrounds in terms of education, occupation, and household size, enabling a comprehensive assessment of water use, access, and governance perceptions at the community level [7, 45]. In addition, key informant interviews were conducted with local government officials, water management authorities, and community leaders, whose institutional roles provided governance-level perspectives on water management and policy implementation within the catchment [52]. These stakeholders were identified through a snowball approach. The data collection strategy was designed to generate statistically valid and contextually grounded insights into water usage patterns, accessibility, and management perceptions among residents, while allowing

triangulation between quantitative household data and qualitative governance-level evidence [14, 19].

3.3 Sample size calculation

The sample size for the household survey was determined to ensure adequate statistical power and representativeness of the population within the Owabi River Catchment [12]. Cochran's sample size formula for categorical data was applied using a 95% confidence level, a margin of error of 5%, and an assumed population proportion of 0.5, which is recommended when the true population proportion is unknown and variability is expected to be high [12, 31]. Based on an estimated household population of approximately 5000 within the catchment, the minimum required sample size was calculated and adjusted to account for potential non-response and incomplete questionnaires [31]. Consequently, a total of 1140 households were targeted and 1015 households finally participated in the study with a participation rate of approximately 89%, which exceeds commonly accepted thresholds for survey reliability in community-based studies [8]. This sample size was considered sufficient to support robust descriptive and inferential analyses of water security and governance variables at the catchment level [14].

3.4 Sampling strategy

Data were collected across four communities (Owabi-400, Kronum-250, Adoato-Adu-manu-200, and Bokankye-200) identified through a stratified sampling approach with strata defined by settlement type (urban, peri-urban, and rural), population size, and distance from water sources, followed by proportionate allocation and systematic random sampling within each stratum to minimise sampling bias [12, 14]. The structured questionnaires collected information on water sources and access, water use practices, governance perceptions, water quality awareness, conservation behaviour, and key socio-economic characteristics of households [8, 14]. Furthermore, field observations were undertaken to document environmental conditions such as encroachment on buffer zones and visible pollution of water bodies, supporting data triangulation and enhancing the validity of the study findings [14]. This was possible through regular site visits at various points along the Owabi River and its tributaries. Photographs provided visual documentation supporting the written observations.

To complement the quantitative data, twelve (12) structured in-depth interviews were conducted with diverse local and national government administrators from different institutions and departments mandated to manage the environmental and water resources among these comprise: municipal spatial planners, water management officials, and traditional leaders. This was to capture institutional and governance perspectives relevant to water security in the catchment. The interviews were also to provide insights concerning the management of the catchment and the Owabi River. Each interview took about 25–35 min and were recorded with the consent of the participants.

3.5 Data analysis

3.5.1 Water security analysis

Quantitative data related to water security were analysed using Stata version 15 [49]. Descriptive statistics, including frequencies, percentages, means, and standard deviations, were used to summarise household socio-demographic characteristics, sources of

water, access patterns, reliability of supply, and perceptions of water quality [8]. Cross-tabulations and Chi-square tests were applied to examine bivariate associations between water security indicators and household characteristics such as settlement type, distance to water sources, and socio-economic status [5]. Observational data collected during community visits, including evidence of pollution, encroachment on buffer zones, and water source conditions, were documented using photographs and field notes to support the interpretation of survey findings [14].

3.5.2 Water governance analysis

Water governance data were analysed using both quantitative and qualitative approaches to capture institutional, participatory, and management dimensions [14]. Descriptive statistics were used to summarise household perceptions of governance effectiveness, community participation, pollution control measures, waste management practices, and water treatment arrangements [8]. To examine determinants of sustainable water resource management, a binary logistic regression model was estimated, with perceived effectiveness of water management coded as effective (1) and not effective (0) [29].

Independent variables included community involvement, water treatment practices, pollution control, and waste management, selected based on theoretical relevance and bivariate screening at $p < 0.25$ [29]. Potential confounding variables such as income, education, and household size were assessed during model specification but excluded due to collinearity or lack of statistical significance [36]. Model adequacy was assessed using the Hosmer–Lemeshow goodness-of-fit test, pseudo- R^2 statistics [13, 40], and Variance Inflation Factor (VIF) tests for multicollinearity [20]. Regression results were reported as odds ratios (ORs) with 95% confidence intervals and significance set at $p < 0.05$.

Qualitative data from key informant interviews were analysed thematically to identify recurring patterns related to institutional coordination, enforcement challenges, and community participation in water governance [10]. Illustrative quotations were used to complement and contextualise the quantitative findings.

3.6 Ethics adherence

Ethical approval was obtained from the University's Ethics Committee—UNISA-CAES HEALTH RESEARCH ETHICS COMMITTEE REC Reference #: 2022/CAES_HREC/072. Thus, ethical considerations were rigorously observed throughout the research process. Participation was voluntary, and informed consent was obtained from all participants. Informed consent was obtained from all participants, ensuring voluntary participation and confidentiality. Furthermore, authorization was sought from the local administrators in the study site.

4 Results

4.1 Demographics of the sample

A total of 1,015 participants took part in the survey. Of these, 472 were male (46.5%) and 543 were female (53.5%). Regarding age distribution, 286 participants (28.2%) were between 18 and 29 years, 312 (30.7%) were between 30 and 39 years, 232 (22.9%) were aged 40–49 years, and 185 (18.2%) were aged 50 years and above. In terms of educational attainment, 198 participants (19.5%) reported having no formal education, 272 (26.8%) had primary education, 369 (36.4%) had secondary education, and 176 (17.3%) had

tertiary education. With respect to occupation, 284 respondents (28.0%) were engaged in farming, 331 (32.6%) were involved in trading, 244 (24.0%) worked as artisans, and 156 (15.4%) were in formal employment. Household size varied among the participants, with 238 households (23.4%) comprising 1–3 members, 489 households (48.2%) comprising 4–6 members, and 288 households (28.4%) comprising seven or more members. In relation to settlement type, 430 participants (42.4%) resided in urban areas, 356 (35.1%) lived in peri-urban areas, and 229 (22.6%) were from rural communities (Appendix 1).

4.2 Water sources, uses and water security around the Owabi river catchment

Information on sources of water within the study communities indicated that 47.2% (522) of the participants had access to pipe-borne water. This was followed by wells, reported by 22.4% (247), mechanised boreholes by 15.0% (166), and boreholes by 14.1% (156). Furthermore, 1.3% of the sampled population noted that the Owabi River is their main source of water in the area (Table 1).

Regarding the location of water sources, 57.5% (635) of participants reported that their primary water source was located within the household. In contrast, 30.2% (334) indicated that their water source was located outside the house, while 12.3% (136) reported that their water source was located within the community.

With respect to the distance of water sources from households, 72.8% (804) of the participants indicated that their main water source was located within 100 m of their

Table 1 Features of water sources used in the Owabi Community

Variable	Frequency	Percent (%)
Source of water in the community		
Borehole	156	14.1
Mechanised Borehole	166	15
Owabi River	14	1.3
Pipe Borne	522	47.2
Well	247	22.4
Location of Source Water		
Within the House	635	57.5
Outside the House	334	30.2
Within the Community	136	12.3
Distance of Source from the House		
Under 100 m	804	72.8
101 m to 300 m	221	20
301 m to 500 m	66	6
501 m to 800 m	11	1
At least 801 m	3	0.3
Condition of Water		
Bad	25	2.3
Fair	307	27.8
Good	773	70
Frequency of Water Flow		
Daily	995	90
Once a Week	19	1.7
Twice a Week	8	0.7
Three Times a Week	83	7.5
Is Owabi River Polluted		
Yes	924	83.6
No	181	16.4

Source: (Fieldwork, 2023)

residence. Distances of 101–300 m were reported by 20.0% (221), while 6.0% (66) reported distances ranging from 301–500 m. Smaller proportions reported distances of 501–800 m, accounting for 1.0% (11), and distances of at least 801 m, reported by 0.3% (3). In relation to the condition of water, 70.0% (773) of the participants described it as good, while 27.8% (307) described it as fair. A smaller proportion, 2.3% (25), reported that the condition of the water was bad (Table 1).

The study also found that water flow daily and this was reported by 90.0% (995) of the study population. For example, water flow three times a week 7.5% (83), while 1.7% (19) of the study population reported that water flow once a week and 0.7% (8) reported that water flow twice a week. Finally, responses on the pollution status of the Owabi River showed that 83.6% (924) of the study population reported that the river was polluted, whereas 16.4% (181) reported that it was not polluted (Appendix 2).

4.3 The state of water in the Owabi river catchment

The study revealed substantial concerns regarding the condition of water in the Owabi River Catchment. Overall, 83.62% (924) of the participants indicated that the Owabi River was polluted, while 16.38% (181) reported that the river was not polluted (Appendix 2). Among participants who acknowledged river pollution, varying perceptions of pollution intensity were reported. High levels of water pollution were reported by 38.28% (423) of the participants, while 21.00% (232) indicated that pollution levels were very high. In addition, 22.81% (252) of sampled population described pollution levels as low, and 4.62% (51) indicated very low levels of pollution. Sources of pollution within the catchment were linked to waste disposal and liquid waste management practices. Liquid waste disposal through soak-away systems accounted for 34.66% (383) of responses, while disposal through drains constituted 31.40% (347). Septic tanks were reported by 26.33% (291) of the participants, open surface disposal by 5.07% (56), and catch pits by 2.53% (28). Regarding health implications associated with water quality, 11.49% (127) of the participants reported experiencing health issues related to water conditions, while 88.51% (978) reported no health-related effects (Appendix 2). Among reported health conditions, diarrhoea was identified by 55.01% (538) of the participants, typhoid fever by 43.05% (421), cholera by 0.72% (7), giardia by 0.41% (4), and dysentery by 0.82% (8). With respect to regulatory compliance, 58.64% (648) of the participants indicated that permits were obtained for relevant waste or water-related activities, while 41.36% (457) reported the absence of permits (Appendix 2). Although 58.64% (648) of the population reported holding permits for waste disposal, 41.36% (457) indicated that they lacked formal authorisation. Interviews with a community member on Atafua regarding the state of water in the Owabi River revealed that:

'The poor water situation, especially the rampant encroachments and pollution levels, compels us to redirect significant resources to management practices. This includes more frequent water treatment cycles and surveillance, which are costly but unavoidable.'

This quotation emphasises the urgent need for effective water management practices as influenced by existing water conditions.

Furthermore, a 59-year-old male Chief from Akyeampomene, expressed his concern:

"It's gotten to the point where you can smell the pollution in the water. My neighbour fell sick last month, and the doctors suspect it's due to the contaminated water supply."

Coming from a traditional leader, this chief's comments carry the weight of his community and highlight the urgent health risks posed by declining water quality. The high prevalence of pollution, coupled with inadequate sanitation measures and health risks, underscores the need for stronger enforcement of environmental laws, enhanced public awareness programs, and investment in sustainable waste management solutions. Without immediate interventions, the deterioration of the Owabi River's water quality could further endanger community health and ecosystem stability.

4.4 Governance and sustainability of the Owabi river catchment

The governance of water resources in the Owabi River Catchment is characterised by limited community participation in decision-making processes, inconsistent enforcement of waste management regulations, and weak coordination among institutions responsible for water management. A substantial proportion of households operate without formal permits for waste disposal, indicating gaps in regulatory compliance and monitoring mechanisms. In addition, pollution control remains a challenge due to inadequate management of domestic wastewater, agricultural activities, and land-use practices within the catchment. These governance challenges collectively affect the sustainable management of water resources in the Owabi River Catchment, as reflected in household-level governance and management indicators. Interview with a 37-year-old NGO Worker from Owabi regarding sustainable practices showed that:

"Deforestation not only affects our environment but also directly impacts our water sources. It's a vicious cycle we're failing to address adequately."

Similarly, a 54-year-old Chief from Sukobri underscored the importance of community involvement and traditional water management practices, stating that:

"Our ancestors knew the land and how to care for it; we should draw on that wisdom and involve the community in every aspect of water management. It's the only way to ensure a sustainable future."

These revelations emphasize the importance afforestation and environmental conservation to protect the water catchments which is the main water source of the community. These could be enhanced by NGO's with experience in environmental management. Furthermore, traditional authorities have a great say in environmental management. The words of community leaders and village chiefs carry authority and respect, highlighting the intrinsic link between community involvement and sustainable water management practices. A 42-year-old female teacher from Adoato-Adumanu further highlight the role of education in conservation, stating that;

"Teaching our children about conservation and responsible land use is crucial. If we do not educate the next generation, these problems will persist."

These revelations from traditional authorities, instructors and community developers offer valuable insights on environmental management, the incorporation of community

Table 2 Sustainable management strategies

Sustainable Management Strategies (effective/not effective)	Category (Reference)	B (Coef.)	SE	OR	95% CI for OR	p-value
Community Involvement	Not Involved (ref=Involved)	2.9	0.17	18.12	[13.04, 25.13]	< 0.001*
Water Treatment	Untreated (ref=Treated)	-0.1	0.19	0.9	[0.63, 1.30]	0.591
Pollution control	Polluted (ref=Not Polluted)	0.28	0.22	1.32	[0.86, 2.03]	0.205
Waste (Waste Management)	Open Drainage (ref=None)	-0.94	0.48	0.39	[0.15, 1.01]	0.052
	Septic Tank	0.37	0.19	1.45	[0.99, 2.11]	0.055
	Bush	-0.03	0.51	0.97	[0.36, 2.61]	0.947
	Proper Drainage	-0.52	0.21	0.59	[0.39, 0.90]	0.015*
Constant		-1.72	0.24	0.18	[0.11, 0.29]	< 0.001*

*($p < 0.05$) indicates attributes with significant implications

and traditional values in natural resource management. It can be argued that education and knowledge transfer as a fundamental pillar for sustainability.

Household responses on payment for river protection and community involvement were captured using binary indicators, and their associations with protection efforts (Appendix 3a, b). The results show a statistically significant association between health concerns and community efforts to protect the Owabi River ($\chi^2 = 23.0782$, $p < 0.001$). Among participants who reported experiencing health problems related to water quality, 76 participated in conservation efforts, while 51 did not. Among participants who did not report health-related concerns, 368 participated in conservation efforts, and 610 did not. A statistically significant association was also observed between institutional support and community participation in river protection activities ($\chi^2 = 7.9983$, $p = 0.005$). Of the participants who reported receiving institutional support, 243 participated in conservation efforts as opposed to 418 participants. In contrast, among participants who reported no institutional support, 201 participated in conservation efforts as opposed to 243 participants (Appendix 3b). Overall, household responses indicate that both health concerns and institutional support are associated with variations in community participation in efforts to protect the Owabi River.

4.4.1 Integrated water management options of the Owabi river catchment

The results showed that integrated water management in the Owabi River Catchment is centred on community involvement and household-level waste management practices, including water treatment, pollution control, and liquid waste disposal methods (Table 2). For example, community involvement emerged as a strong predictor, with individuals who were not involved in community efforts being significantly less likely to support sustainable water practices ($B = 2.90$, $p < 0.001$, $OR = 18.12$, 95% CI: [13.04, 25.13]) (Table 2). This suggests that active community participation increases the odds of sustainable water management by over 18 times, emphasising the importance of grassroots engagement in environmental governance.

The findings reinforce the value of inclusive governance and adequate sanitation infrastructure in advancing water sustainability. However, the effectiveness of these strategies varies across communities, with some approaches producing more consistent outcomes than others.

5 Discussion

5.1 Water security in the Owabi river catchment

This study revealed that water security in the Owabi River Catchment is primarily constrained by water quality degradation, pollution pressures, and associated health risks, despite the presence of multiple water sources. Although households access water from pipe-borne systems, wells, boreholes, and the Owabi River, concerns about pollution indicate that availability alone does not guarantee water security [42]. This finding aligns with the broader understanding of water security as encompassing not only physical access but also water quality and safety for human and ecosystem use [59]. Studies also revealed that the widespread perception of river pollution and reported exposure to waterborne diseases underscore the close linkage between environmental degradation and public health within the catchment [7, 39]. Similarly, studies in Ghana and other sub-Saharan African contexts have documented how untreated domestic wastewater, agricultural runoff, and urban land-use changes contribute to declining surface water quality and increased health vulnerabilities [19, 27, 43]. The Owabi Catchment reflects these dynamics, where pollution compromises both household water use and ecological integrity, reinforcing the multidimensional nature of water insecurity. Accessibility-related findings further indicated that proximity to water sources and frequent water flow do not necessarily translate into secure water access when water quality is poor. In contrast, the WHO / UNICEF [62] emphasis on time, distance, emergency standards as essential safety criteria for accessibility. This supports earlier studies which argued that water security cannot be assessed solely on distance or availability indicators, but must incorporate water safety and reliability dimensions (FAO, 2019, [30]). In the Owabi Catchment, households located close to polluted water sources remain exposed to health risks, demonstrating that infrastructure alone is insufficient without effective source protection. Local coping strategies, including household-level water treatment and reliance on alternative sources, illustrate community resilience in the face of water insecurity. Similar adaptive responses have been reported in other Ghanaian catchments, where households adopt short-term solutions to compensate for weak environmental protection and service provision [50]. However, while these strategies reduce immediate vulnerability, they do not address the underlying drivers of water insecurity, particularly pollution and land-use pressures. This suggests that long-term water security in the Owabi River Catchment depends on catchment-scale interventions that integrate water quality protection, land management, and public health considerations.

5.2 Water governance in the Owabi river catchment

The findings further demonstrate that water governance in the Owabi River Catchment is characterised by institutional fragmentation, weak enforcement, and limited depth of community participation. Although Ghana's water governance framework is grounded in principles of decentralisation and Integrated Water Resources Management, its effectiveness at the catchment level remains constrained. This mirrors earlier assessments, which highlight coordination gaps among regulatory institutions and local authorities as persistent challenges in Ghana's water governance system [52, 35]. Furthermore, community participation emerged as an important but uneven element of water governance within the catchment. While some households engage in river protection and management activities, participation is often informal and lacks sustained institutional support.

This finding is consistent with studies showing that participation in many African water governance contexts remains largely consultative rather than empowering, limiting its influence on enforcement and decision-making processes [11, 47, 53]. In the Owabi Catchment, limited community involvement reduces collective capacity to prevent pollution and enforce local environmental norms.

Effluent waste management practices further expose governance weaknesses affecting water sustainability. The coexistence of regulated systems such as septic tanks with informal disposal methods indicates gaps in regulatory oversight, infrastructure provision, and compliance monitoring. Similar governance-related challenges have been documented in other urban and peri-urban catchments in Ghana, where inadequate sanitation governance exacerbates surface and groundwater pollution [19, 45]. These findings suggest that ineffective waste governance directly undermines water resource protection in the Owabi Catchment. Despite these challenges, the study also highlights locally driven solutions that contribute to water governance and sustainability. Community-led conservation efforts, household waste management practices, and engagement with municipal and traditional authorities demonstrate the potential for adaptive and participatory governance approaches. This aligns with polycentric governance perspectives, which emphasises the role of multiple actors and local initiatives in managing common-pool resources such as water [46, 47, 54]. Strengthening institutional coordination, improving enforcement capacity, and formally integrating local initiatives into governance frameworks are therefore critical for improving water governance outcomes and enhancing the sustainability of the Owabi River Catchment.

6 Conclusion

Substantial water security challenges, particularly widespread river pollution, declining water quality, and health risks linked to domestic wastewater disposal and land-use pressures, has transform the biodiversity of the Owabi River catchment and influence the livelihood and wellbeing of the surrounding communities. Governance-related challenges characterised by weak regulatory enforcement, limited institutional coordination, and predominantly passive forms of community participation in water management decision-making persist. Water security concerns are driven largely by pollution and access constraints, governance failures, and particularly limited community engagement. Hence, strengthening participatory governance, improving pollution control, and integrating technological solutions is recommended for the sustainability of the Owabi River Catchment.

Appendix

Appendix 1: Demographics of Samples

Variable	Frequency	Percent (%)
Sex		
Male	472	46.5
Female	543	53.5
Age group (years)		
18–29	286	28.2
30–39	312	30.7

Variable	Frequency	Percent (%)
40–49	232	22.9
50 and above	185	18.2
Education level		
No formal education	198	19.5
Primary	272	26.8
Secondary	369	36.4
Tertiary	176	17.3
Occupation		
Farming	284	28
Trading	331	32.6
Artisan	244	24
Formal employment	156	15.4
Household size		
1–3	238	23.4
4–6	489	48.2
7 and above	288	28.4
Settlement type		
Urban	430	42.4
Peri-urban	356	35.1
Rural	229	22.6

Appendix 2: Analysis of Water Pollution Sources in the Owabi River Catchment. Source: (Fieldwork, 2023)

Pollution	Freq	Per cent
Yes	924	83.62
No	181	16.38
Pollution		
Yes	924	83.62
No	181	16.38
Rate of water pollution		
Very low	51	4.62
Low	252	22.81
High	423	38.28
Very high	232	21
Health Implication		
Yes	127	11.49
No	978	88.51
Health issues		
Diarrhoea	538	55.01
Typhoid fever	421	43.05
Cholera	7	0.72
Giardia	4	0.41
Dysentery	8	0.82
Dispose of Liquid		
Open Surface	56	5.07
Drains	347	31.4
Soak away	383	34.66
Catch pit	28	2.53
Septic tank	291	26.33
Permit		
Yes	648	58.64
No	457	41.36

Appendix 3a: Analysis of Health Implications and Community Efforts. Source: (Fieldwork, 2023)

Health Implication	Effort		Total	Pearson chi2(	Pr
	Yes	No			
Yes	76	51	127	23.0782	0.000
No	368	610	978		
Institution				7.9983	0.005
Yes	243	418	661		
No	201	243	444		

Appendix 3b: Binary Logistic Regression of Management of Water Sources. Source: (Fieldwork, 2023)

Management of Water Sources (effective/not effective)	Category	B (Coef.)	SE	OR	95% CI for OR	p-value
Community Involvement	Involved (ref=No)	−2.92	0.17	0.05	[0.038, 0.076]	< 0.001
Water Treatment	Treated (ref=Untreated)	0.35	0.20	1.42	[0.96, 2.10]	0.08
Pollution Control	Not Polluted (ref=Polluted)	−0.24	0.22	0.79	[0.51, 1.22]	0.29
Health Concerns	No (ref=Yes)	−0.83	0.25	0.44	[0.27, 0.71]	0.00
Waste Management	Soakaway (ref=None)	1.01	0.49	2.74	[1.04, 7.18]	0.04
	Septic Tank	1.20	0.49	3.32	[1.26, 8.66]	0.02
	Bush	0.72	0.69	2.06	[0.53, 7.86]	0.30
	Proper Drainage	0.25	0.50	1.28	[0.48, 3.42]	0.62
Governance	Yes (ref=No)	−0.21	0.17	0.81	[0.58, 1.14]	0.23
Economic Factors	Yes (ref=No)	0.12	0.17	1.13	[0.82, 1.57]	0.46
Farming & Chemical Use	No (ref=Yes)	0.39	0.18	1.48	[1.04, 2.10]	0.03
Constant		0.90	0.59	2.46	[0.77, 7.77]	0.13

Author contributions

Agyeman, N. K., Data collection and conceptualization, Tantoh, H. B., helped supervise the project, contributed to the final version of the manuscript, provided critical feedback and helped shape the research, analysis and manuscript. Kamika, I., helped supervise the project, provided critical feedback, and contributed to shaping the research and analysis.

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Data availability

We do not have consent to share the generated data. However, it may only be provided upon reasonable request.

Declarations

Ethical approval and consent to participate

Ethical clearance was obtained from the University of South Africa-College of Agriculture and Environmental Science Health Research Ethics (UNISA-CAES HEALTH RESEARCH ETHICS). This study was approved by the Research Ethics Committee of the University of South Africa-College of Agriculture and Environmental Science, Health Research Ethics-UNISA-CAES HEALTH RESEARCH ETHICS REC—Reference #: 2022/CAES_HREC/072. It is further certified that the study was conducted in accordance with the National Health Research Ethics Council – South Africa (Registration # REC-170616—051). Informed consent was sought from all the participants.

Consent to publish

Not Applicable.

Competing Interest

The authors declare no competing interests.

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